Explaining Gas Price Dynamics in Italy and Turkey

Business, Economic and Financial Data Project

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Abstract

This study examines the determinants of natural gas price dynamics in Italy and Turkey through an explanatory rather than forecasting-oriented approach. Using a set of econometric models such as linear regression, SARIMA, ARIMAX and Generalized Additive Models (GAM), we evaluate how internal time-series structures and external economic and climate-related factors contribute to price variation. The analysis relies on biannual gas price data combined with Brent crude prices, Henry Hub natural gas prices, European temperature anomalies, country-specific structural shocks, and trend indicators.

For Italy, the results show that models incorporating nonlinear trend components provide the best explanatory power, with the GAM specification achieving the strongest overall performance across AIC, MSE and R^2 metrics. This suggests that Italian gas prices follow a smooth but non-constant trend, further influenced by the 2021 price surge. In contrast, the Turkish series is dominated by a strong linear downward trend and pronounced seasonality, leading the seasonal ARIMAX model, which combines external regressors with a seasonal AR structure, to outperform alternative specifications.

Overall, the comparison reveals that the two markets exhibit structurally different price dynamics: Italy is best described by models capturing nonlinear adjustments, while Turkey requires formulations that account for seasonal persistence and deter-

ministic shocks. These findings highlight the importance of choosing model structures that reflect the underlying economic and temporal characteristics of each market.

1 Introduction

Natural gas prices are an important component of economic activity, affecting households, production costs, and energy markets. In Europe, differences in market structures raise the question of whether gas prices across countries respond to similar or distinct economic mechanisms. This study addresses this issue by comparing Italy and Turkey, two countries with clearly different gas price dynamics.

1.1 Motivation

Recent volatility in European energy markets, particularly the sharp price increase around 2021, highlights the need for an explanatory analysis of gas price movements. Italy experienced a sudden and pronounced price surge, suggesting structural breaks and nonlinear behavior. Turkey, by contrast, shows a persistent downward trend with regular seasonality. These contrasting patterns motivate a comparative study of the factors driving gas prices in each country.

1.2 Problem Definition

The main research question is: *Which internal time-series characteristics and external economic or climate-related factors explain changes in natural gas prices in Italy and Turkey?* The analysis examines the roles of trends, seasonality, structural shocks, and external variables such as Brent oil prices and temperature anomalies. The focus is explanatory, aiming to identify which modeling structures best reflect the underlying price formation mechanisms.

2 Data Description

The study examines gas prices for household consumers in Italy and Turkey using biannual data from the first semester of 2007 to the first semester of 2025. According to the Eurostat database (updated as of October 15, 2025) gas prices for household consumers refer to consumption in Band 2, corresponding to annual consumption from 20 GJ to 199 GJ, excluding taxes and levies. For both time series, the considered unit of measure is Euro. These series reflect domestic pricing mechanisms and country-specific energy market characteristics. Several external regressors are included to account for broader influences. We arbitrarily chose three time series that were thought to bring information to our models about three important factors that are well known to contribute to the gas prices determinations.

An international gas market benchmark, the Henry Hub natural gas spot price, measured in Dollars per million Btu. This price is determined by the supply and demand for natural gas at a major pipeline hub in Erath, Louisiana, and is used for spot market trades of natural gas for immediate delivery. The monthly times series was provided by the U.S. Energy Information Administration, EIA (updated as of November 19, 2025).

A proxy for global energy market conditions, the Brent crude oil prices, measured in USD per barrel. The term Brent refers to a type of crude oil extracted mainly in the North Sea and its price is known to be historically correlated with oil's. The monthly time series we used was provided by the U.S. Energy Information Administration, EIA (updated as of November 19, 2025).

And a metric for climate-related abnormalities, the monthly time series of the land average temperature anomaly in Europe, that are known to affect the energy demand. The data was provided by the National Centers for Environmental Information, NCEI.

All the just described variables are harmonized to a biannual (six-month) frequency, covering the time interval from the first semester of 2007 to first semester of 2025.

In addition to these external regressors, we also made use of some synthetic time related variables: Italy's specification includes two trend components to allow for a structural change around 2021, associated with the sharp post-pandemic price surge. Turkey's series is characterized by a single linear trend reflecting its persistent downward trajectory. In addition, shock dummy variables are introduced to capture major structural events, namely the 2021–2022 price spike in Italy and the 2012–2016 policy adjustment period in Turkey.

2.1 Exploratory Plots: Gas Price Series

A visual inspection of the gas price series highlights substantial differences between the two countries. Italy's prices remain relatively stable until 2020, followed by a sharp and unprecedented increase in 2021. Turkey's prices instead display a smoother pattern, dominated by a long-term downward trend with moderate volatility and mild seasonality. These contrasting behaviors suggest the need for different modeling approaches across countries.

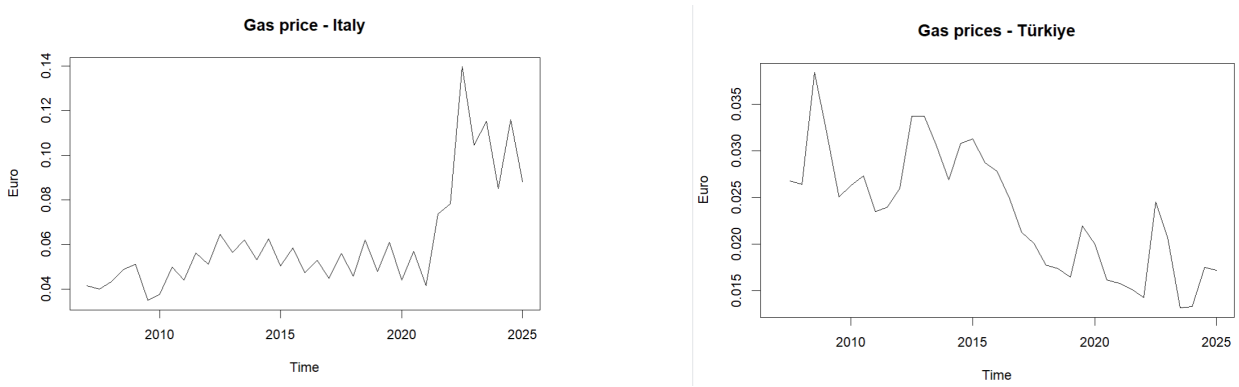

Figure 1: Biannual natural gas prices for Italy and Turkey (2007–2025).

2.2 Preliminary Relationships Among Variables

As an exploratory step, pairwise correlations among gas prices and selected external variables are examined. Figure 2 provides a descriptive overview of these relationships prior to model-based analysis.

Italy's gas prices exhibit a strong positive correlation with temperature anomalies, suggesting a close link between climate conditions and consumption-driven price movements. In

contrast, Turkey's gas prices display a strong negative correlation with temperature anomalies, indicating a different demand-response mechanism. Turkey's prices are also positively correlated with international energy benchmarks, particularly Brent and Henry Hub prices, whereas Italy's correlations with these global indicators appear weaker.

These correlations are interpreted strictly as descriptive evidence. They motivate the inclusion of external regressors in the econometric models, while their explanatory relevance is formally evaluated in the Results section.

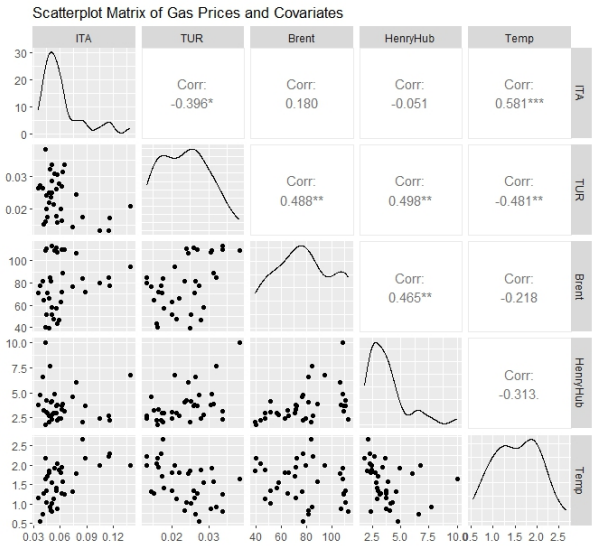

Figure 2: Correlation matrix among gas prices and selected external variables (biannual data).

3 Modeling Hypotheses and Main Research Question

The objective of this study is to identify the main factors driving the evolution of residential natural gas prices in Italy and Turkey, and to assess which econometric models best explain these dynamics. Adopting an explanatory rather than predictive perspective, the analysis focuses on evaluating the role of economic, climatic, and structural variables, as well as on comparing alternative modeling frameworks in terms of their explanatory adequacy.

The first set of hypotheses concerns the impact of external variables. For Italy, global energy market conditions are expected to play a significant role, implying a positive relationship between Brent crude oil prices and gas prices. Temperature anomalies are assumed to influence demand-related seasonal patterns, while the sharp price increase observed in 2021–2022 is captured through a structural shock dummy. For Turkey, in contrast, gas prices are hypothesized to follow a persistent downward trend with limited seasonal variation. While temperature anomalies may still affect consumption. Because Turkey's household gas prices are set differently, Brent might matter less. But we don't include Brent in our Turkey models, so we treat this as background motivation, not something we tested in our results.

A second set of hypotheses relates to model performance. For Italy, more flexible models such as ARIMAX and GAM are expected to outperform simple linear specifications,

given the presence of nonlinear behavior and a major structural break. For Turkey, where price dynamics appear smoother, models incorporating deterministic trends and seasonal components are considered suitable. Nonetheless, by combining these features with external regressors. We expect ARIMAX to beat the simpler time-series models, especially for Turkey. For Italy, because of the big post-2021 break, we expect a more flexible model like a GAM to fit best.

These hypotheses motivate the central research question of the study:

> *Which external factors most effectively explain the evolution of natural gas prices in Italy and Turkey, and which econometric models best capture these dynamics?*

4 Models Adopted

This section outlines the econometric models used to analyze residential natural gas prices in Italy and Turkey. Adopting an explanatory perspective, the modeling strategy progresses from simple linear specifications to more flexible time-series frameworks capable of capturing seasonality, serial dependence, and nonlinear dynamics.

The analysis follows a sequential approach. Linear regression models are first estimated to provide an interpretable baseline and to assess the role of trends, external covariates, and structural shocks. Seasonal ARIMA models are then introduced to capture temporal dependence and seasonal patterns. The ARIMAX specifications extend these models by incorporating external regressors, while Generalized Additive Models (GAM) allow for nonlinear relationships and smoothly evolving trends. This progression enables a systematic comparison of alternative explanatory structures.

4.1 Linear Models

Linear models serve as the baseline framework for both countries, offering a transparent interpretation of gas price dynamics. Multiple specifications are considered to evaluate the relative importance of deterministic trends, external variables, and structural shocks.

4.1.1 Italy

For Italy, linear models are used to establish a first-order explanation of price movements and to assess whether simple deterministic structures are sufficient. Given the pronounced change in price behavior after 2021, a single linear trend fails to capture the full evolution of the series. To address this, a time-based specification decomposes the trend into two components: one capturing the relatively stable pre-2021 period, and a second accounting for the sharp post-2021 price increase. This structure partially accommodates the structural break while remaining within a linear framework.

An alternative specification replaces time-derived features with external regressors, including Brent crude oil prices, European temperature anomalies, and a shock dummy. These models focus on economically meaningful drivers and temporal structure and serve as the baselines for subsequents ARIMAX extensions.

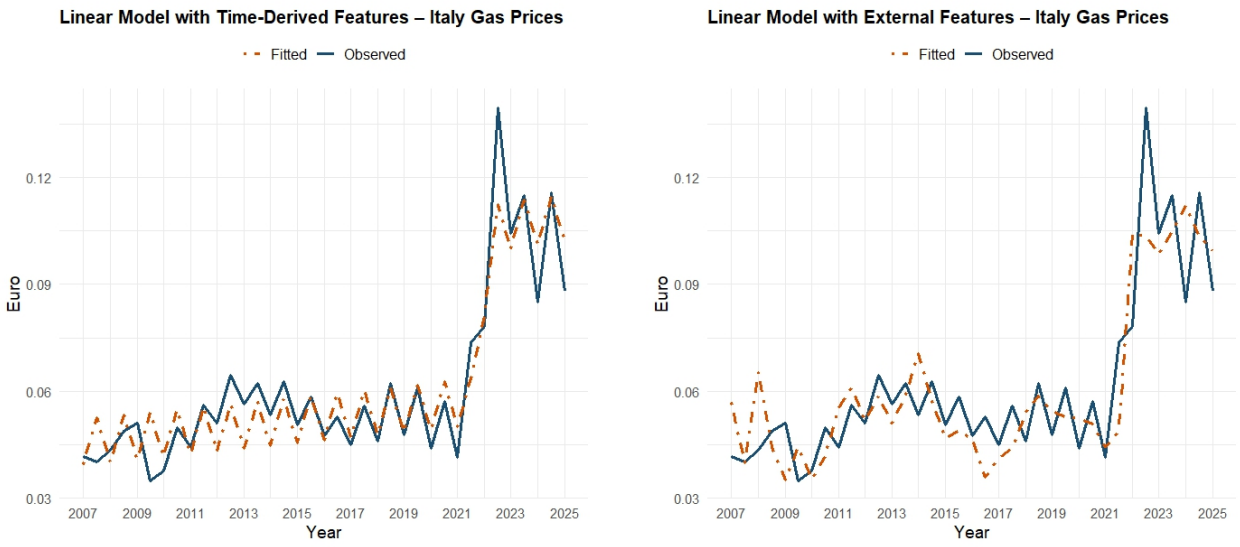

Figure 3: Italy – Linear models fitted values

Figure 3 shows that both linear specifications capture the general price level and medium-term fluctuations. The obtained linear models have great explanatory performance overall, despite not being able to fully reproduce the magnitude of the post-2021 price surge. However, their residuals analysis, displayed in Figure 4, highlights a significant autocorrelation, which breaks one of the fundamental assumptions of the linear model: the residuals' independence. The diagnosis leads us to choose a class of models capable of capturing temporal dependencies, the ARIMA models.

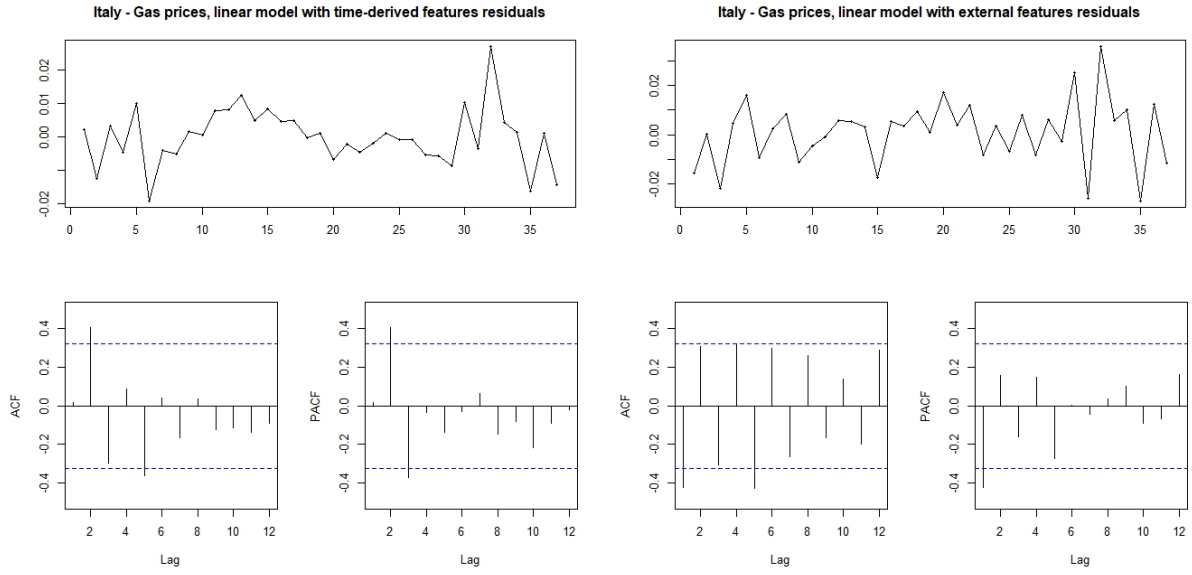

Figure 4: Italy – Residuals analysis of the linear models

4.1.2 Turkey

For Turkey, linear models constitute a more appropriate baseline than in the Italian case, reflecting the smoother and more regular evolution of gas prices over time. The Turkish series does not exhibit an abrupt structural break, but instead follows a persistent downward trend with moderate seasonal variation. As a result, linear specifications are able to capture a substantial share of the underlying price dynamics.

Several linear formulations are considered to evaluate the role of deterministic trends, structural shocks, and climatic conditions. The final specification includes a trend component to model the long-run decline in prices, a shock dummy to account for the exceptional deviation that occured in the 2012-2015 period, and temperature anomalies to reflect demand-related fluctuations. This model serves as the linear benchmark and provides the baseline for the subsequent ARIMAX analysis.

Figure 5 compares the fitted values of the final linear model with the observed Turkish gas prices. The model successfully reproduces the overall downward trajectory of the series and captures medium-term movements. However, short-term fluctuations are not fully captured, indicating that additional dynamic components are required to fully describe the data-generating process.

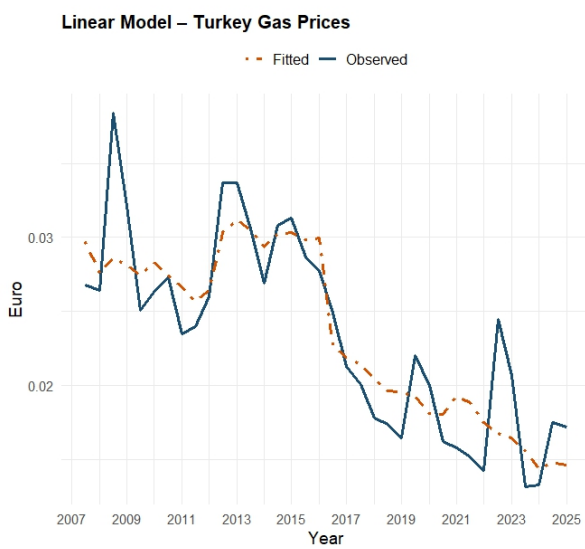

Figure 5: Turkey – Linear model fitted values

Moreover, once more, as displayed in Figure 6, the residual independence assumption of the linear model is broken, due to some temporal dependencies among the residuals. This motivates the extension to models incorporating autoregressive and seasonal structures in the following sections.

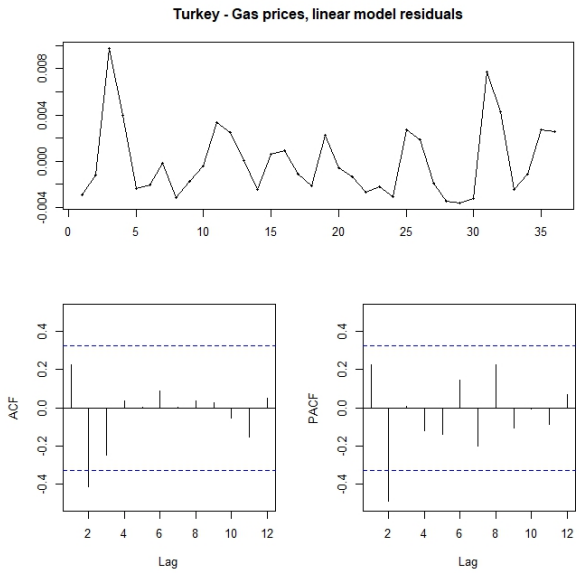

Figure 6: Turkey – Residuals analysis of the linear model

4.2 ARIMA and SARIMA Models

To complement the linear specifications, purely time-series models are considered to evaluate whether internal temporal dynamics alone can explain gas price movements. ARIMA and SARIMA models are used to capture short-run dependence and seasonal patterns without relying on external explanatory variables. This step provides a benchmark and helps motivate the introduction of external regressors in the subsequent ARIMAX specifications.

4.2.1 Italy

For Italy, a pure time-series approach is first explored to assess whether internal dynamics alone are sufficient to explain gas price movements. The inspection of the autocorrelation and partial autocorrelation functions suggests the presence of short-term dependence and pronounced seasonal behavior. As an initial step, a non-seasonal ARIMA(0,0,1) model is considered, capturing the dominant moving-average structure observed in the data.

However, significant seasonal patterns remain, motivating the inclusion of seasonal differencing. This leads to the specification of a SARIMA(0,0,1)(0,1,0)$_2$ model, which accounts for biannual seasonality. Diagnostic checks indicate that residual autocorrelation is largely removed, suggesting that the model adequately captures the internal temporal structure of the series.

Figure 7 compares the fitted values of the SARIMA model with observed Italian gas prices. While the model reproduces the general seasonal fluctuations and medium-term movements, it struggles to fully capture the sharp price surge observed after 2021. This limitation highlights the inability of purely time-driven models to explain major market disruptions and motivates the introduction of external explanatory variables in subsequent ARIMAX specifications.

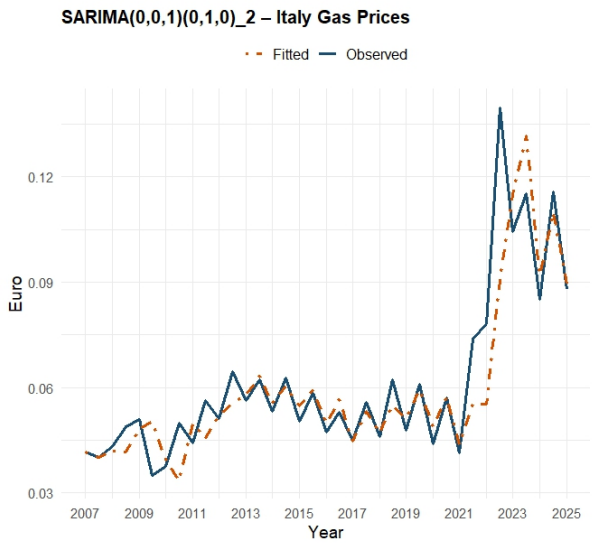

Figure 7: Italy – SARIMA(0,0,1)(0,1,0)$_2$ fitted values

4.2.2 Turkey

For Turkey, time-series models are estimated to capture serial dependence and seasonal structure directly from the gas price series. The initial inspection of the autocorrelation and partial autocorrelation functions suggests a strong non-stationary behavior consistent with a declining trajectory, leading to a first-difference specification. As a starting point, an ARIMA(0,1,0) model is considered to remove the dominant trend component.

Residual diagnostics indicate that the remaining dependence is mainly seasonal. To address this, seasonal components are introduced and alternative SARIMA specifications are evaluated. The final selection is based on improved information-criterion performance, leading to the SARIMA(0,1,0)(1,0,0)$_2$ model, which captures biannual seasonality.

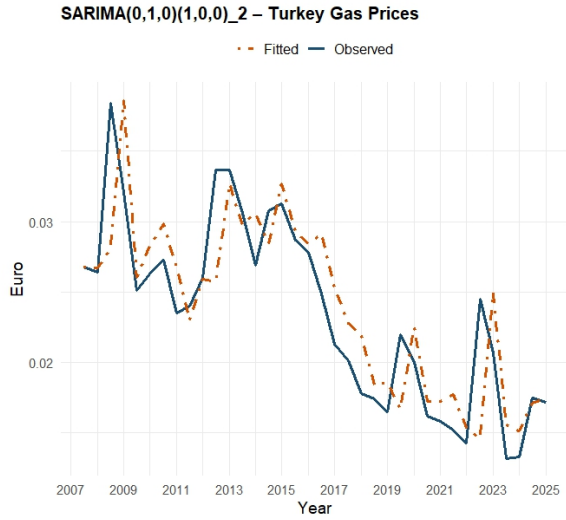

Figure 8: Turkey – SARIMA(0,1,0)(1,0,0)$_2$ fitted values

Figure 8 shows that the SARIMA model successfully tracks the long-run downward trend and reproduces part of the medium-term movements in Turkish gas prices. Nevertheless, some short-term fluctuations remain unexplained, indicating the limits of purely time-driven models and motivating the inclusion of external explanatory variables in the subsequent ARIMAX analysis.

4.3 ARIMAX Models

The ARIMAX framework extends the previous time-series models by explicitly incorporating explanatory variables into the dynamic structure of the series. By combining autoregressive components with external or time-derived regressors, ARIMAX models allow for a joint assessment of internal temporal dependence and structural drivers of gas prices. This section evaluates whether the inclusion of such information leads to a meaningful improvement over purely time-driven specifications.

4.3.1 Italy

For Italy, two alternative ARIMAX specifications are estimated to assess the contribution of additional explanatory information beyond internal time-series dynamics. The two ARIMAX specifications use the linear models described in chapter 4.1.1. as baselines and aim at capturing the residuals time dependencies structure through an ARIMA component.

The first specification incorporates external covariates, including temperature anomalies, Brent crude oil prices, and a structural shock dummy to model the underlying structure of the time series and uses an AR(1) to describe shorter-term fluctuations. While this approach improves upon purely time-driven models, residual diagnostics suggest that the explanatory gains remain limited.

A second specification focuses on time-derived regressors, including segmented trend components, deterministic seasonality, and a shock dummy to capture the main structure of the time series and a SARIMA(0,0,0)(1,0,0)$_2$ to model the tinier autocorrelated components. This formulation yields a substantial improvement in model fit and produces residuals consistent with white noise behavior. The resulting ARIMAX model successfully captures both gradual price movements and the pronounced structural break observed after 2021.

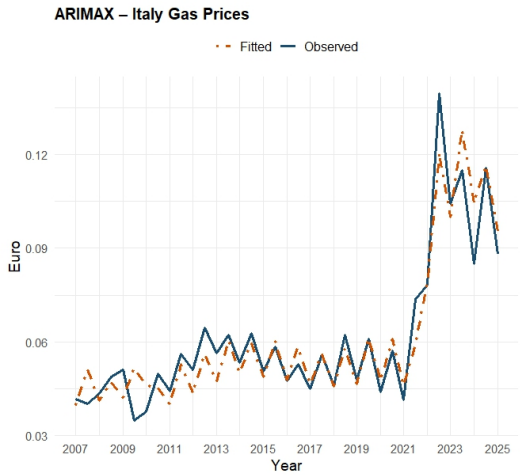

Figure 9: Italy – ARIMAX model with time-derived regressors fitted values

Figure 9 shows that the selected ARIMAX specification provides the most accurate representation of Italian gas price dynamics among the considered models. Compared to SARIMA and linear benchmarks, the inclusion of time-based structural features substantially enhances explanatory performance.

4.3.2 Turkey

For Turkey, the ARIMAX specification builds directly on the final linear benchmark by introducing a dynamic seasonal component. The model includes a deterministic trend, a shock dummy, and temperature anomalies as regressors, and adds a seasonal autoregressive term to account for remaining dependence at the biannual frequency. Residual diagnostics indicate that the seasonal dependence is effectively removed and the residuals behave as white noise, suggesting that the model captures the main systematic structure of the series.

Unlike the pure ARIMA models used for Turkey, where differencing is needed to eliminate the strong downward trend and ensure stationarity, this step is unnecessary in the ARIMAX framework. In ARIMAX, long-run trends and structural movements are directly accounted for through the included external regressors, allowing the autoregressive seasonal term to focus solely on modeling residual short-term dependence and seasonal patterns.

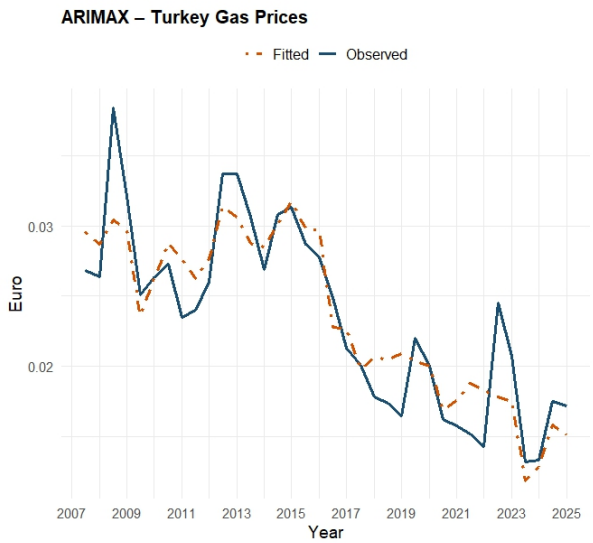

Figure 10: Turkey – ARIMAX model with external regressors fitted values

Figure 10 shows that the fitted values closely follow the observed downward trajectory and medium-term movements, supporting ARIMAX as an adequate explanatory model for Turkish gas price dynamics.

4.4 GAM Models

Generalized Additive Models (GAM) are used to allow for nonlinear relationships between gas prices and selected explanatory variables, while preserving interpretability. In this study, GAMs are employed to assess whether nonlinear time patterns provide additional explanatory power beyond linear and ARIMAX specifications.

4.4.1 Italy

For Italy, Generalized Additive Models are estimated to capture potential nonlinear patterns in gas price dynamics that may not be fully explained by linear or ARIMAX specifications. In particular, segmented time trends are modeled using smooth functions to allow for gradual changes and nonlinear adjustments over different periods.

Several GAM formulations are explored by combining smooth trend components with deterministic seasonality and a structural shock dummy. Initial specifications employ spline-based smoothers, while alternative formulations use local regression smoothers. The latter provide a more flexible representation of trend behavior and result in a substantial improvement in model fit, as indicated by a lower information criterion.

The final GAM specification employs local smoothers for both time-derived trend components, together with seasonal and shock dummies. Figure 11 shows that this model closely tracks observed Italian gas prices, including the sharp increase after 2021. Residual diagnostics indicate no remaining autocorrelation, suggesting that the nonlinear time structure is effectively captured. Given its superior fit, this specification is retained as the representative GAM model for Italy, while alternative GAM formulations are omitted for brevity.

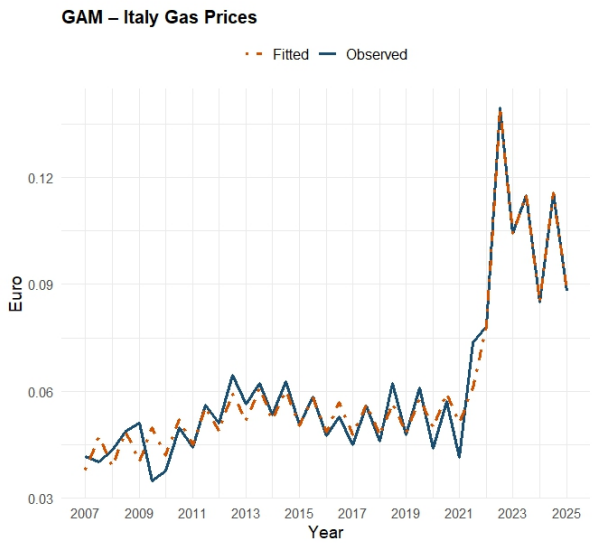

Figure 11: Italy – Generalized Additive Model (GAM) fitted values

4.5 Turkey

Generalized Additive Models are also estimated for Turkey to examine whether allowing for nonlinear effects in the time trend or temperature anomalies improves the explanatory power of the model. Several GAM specifications are considered, combining smooth and local polynomial functions for the trend and temperature variables, together with a structural shock dummy.

For Turkey, the GAM has a higher AIC than both the linear model and the ARIMAX model. So adding nonlinear terms does not improve the fit in our results. This suggests that the Turkish gas price series is characterized by a relatively smooth and gradual evolution, which can be adequately captured by linear and ARIMAX specifications without the need for additional nonlinear flexibility.

Given the absence of a clear improvement in model fit, GAM models are not retained for Turkey, and the subsequent analysis focuses on linear and ARIMAX approaches.

5 Results

This section presents the empirical results of the explanatory models estimated for Italy and Turkey. Model performance is evaluated using multiple complementary criteria: the Akaike Information Criterion (AIC), Mean Squared Error (MSE), and the adjusted R^2. Lower AIC and MSE values indicate better fit, while higher R^2 and adjusted R^2 values reflect stronger explanatory power. All measures are computed on in-sample fitted values to ensure comparability across specifications.

5.1 Italy

Table 1 reports the results for the Italian gas price series. Among linear specifications, the model with time-derived features outperforms the linear model based only on external covariates, highlighting the importance of structural time components. Introducing temporal dependence through SARIMA further improves model performance relative to simple linear benchmarks, confirming the presence of serial correlation and seasonality in the Italian series. ARIMAX models provide additional gains by jointly capturing external regressors and autoregressive dynamics. In particular, the specification combining time-derived covariates with seasonal autoregressive components achieves a clear improvement in AIC and overall fit. The best in-sample performance is obtained by the Generalized Additive Model (GAM), which records the lowest AIC and MSE values, indicating that allowing for nonlinear trend components offers the most accurate explanation of Italian gas price dynamics.

Table 1: Model comparison results for Italy

Model	AIC	MSE	R^2	Adj. R^2
Linear model (time-based)	-235.52	7.28e-05	0.8701	0.8491
Linear model (external covariates)	-208.08	1.70e-04	0.6961	0.6685
SARIMA(0,0,1)(0,1,0)$_2$	-208.23	1.26e-04	0.7756	0.7692
ARIMAX (external covariates)	-213.85	1.37e-04	0.7552	0.7246
ARIMAX (time-based covariates)	-241.45	5.79e-05	0.8967	0.8760
GAM	-261.94	2.36e-05	0.9579	–

5.2 Turkey

Table 2 summarizes the results for Turkey. The baseline linear model already fits well. In our in-sample metrics, pure SARIMA models do not improve over this linear benchmark (they have higher AIC and MSE). Among the SARIMA specifications we tried, SARIMA(0,1,0)(1,0,0)$_2$, is the best, but the main improvement comes from ARIMAX, which combines trend/shock regressors with a seasonal AR term. The inclusion of external information through ARIMAX leads to a further improvement in model performance. The ARIMAX specification combining trend, temperature anomalies, and structural shocks with seasonal autoregressive dynamics achieves the lowest AIC and MSE and the highest R^2 among the considered models. By contrast, the GAM does not outperform the best linear or ARIMAX specifications, indicating that additional nonlinear flexibility does not provide meaningful explanatory gains for the Turkish series.

Table 2: Model comparison results for Turkey

Model	AIC	MSE	R^2	Adj. R^2
Linear model	-303.11	9.78e-06	0.7635	0.7330
SARIMA(0,1,0)(1,0,0)$_2$	-287.11	1.37e-05	0.6694	0.6597
ARIMAX	-308.98	7.75e-06	0.8126	0.7814
GAM	-298.51	8.34e-06	0.7983	–

6 Discussion

This section interprets the empirical results by linking model performance to the structural characteristics of the Italian and Turkish natural gas markets. Consistent with the explanatory focus of the study, the discussion emphasizes economic interpretation rather than predictive accuracy.

6.1 Italy

The results for Italy indicate that gas price dynamics are strongly influenced by nonlinear behavior and structural changes. The superior performance of flexible models reflects the presence of abrupt price movements, particularly after 2021, which challenge linear and purely autoregressive specifications. Models that restrict trend dynamics to fixed functional forms are unable to fully accommodate these regime shifts.

The strong explanatory power of the GAM highlights the importance of allowing nonlinear trend components in markets exposed to international energy shocks. While ARIMAX models benefit from incorporating external variables related to global market conditions and exceptional disruptions, their performance remains inferior to that of the GAM. This suggests that capturing nonlinear temporal dynamics is crucial for explaining Italian gas price movements.

6.2 Turkey

In contrast, the Turkish gas price series exhibits a smoother evolution characterized by a persistent downward trend and moderate seasonality. This structure explains why relatively simple models already achieve strong explanatory performance. The absence of pronounced nonlinearities reduces the need for highly flexible specifications.

The inclusion of external variables through ARIMAX provides additional explanatory gains by accounting for short-term deviations around the long-run trend. However, the limited improvement observed for the GAM indicates that nonlinear modeling offers little added value in the Turkish context. Overall, Turkish gas prices appear to be primarily driven by deterministic trend and seasonal mechanisms.

6.3 Cross-Country Implications

The comparison between Italy and Turkey highlights the importance of adapting model complexity to country-specific price dynamics. Flexible models are essential when prices are affected by structural breaks and nonlinear adjustments, as in Italy. Conversely, more parsimonious specifications are sufficient when price movements follow stable and predictable patterns, as observed for Turkey.

6.4 Limitations

The analysis is subject to several limitations. The biannual frequency of the data may conceal short-term fluctuations, and the set of external variables does not explicitly capture policy or

regulatory interventions. Moreover, the focus on in-sample explanatory performance excludes out-of-sample validation.

7 Conclusion

This study has examined the determinants of residential natural gas prices in Italy and Turkey from an explanatory perspective, with the aim of identifying which factors and econometric models best describe observed price dynamics. Applying a common modeling framework highlights clear differences in both price behavior and model performance across the two countries. For Italy, gas prices are strongly shaped by nonlinear trends and structural changes, particularly following major market disruptions. Models that allow for flexible trend dynamics and incorporate external market information provide a markedly better explanation of Italian price movements than simpler linear or purely time-series specifications. By contrast, Turkish gas prices display a smoother evolution dominated by a persistent long-term trend and moderate seasonality. In this context, parsimonious models focusing on deterministic components and selected external variables perform well, while additional nonlinear flexibility yields limited gains. Overall, the results underline the importance of aligning model complexity with country-specific price structures to achieve meaningful explanatory insight into natural gas price dynamics.

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